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Application of the Optical Fiber Diameter Analyzer for Assessing Cotton Fiber Ribbon Width

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ABSTRACT

Existing methods for measuring fineness are time-consuming, tedious, and impractical to evaluate many samples. For this reason, the Textile Industry has sought a rapid, accurate, and reliable instrument to determine fiber fineness. Recently, the Optical Fiber Diameter Analyzer (OFDA) was introduced for rapid and precise evaluation of the diameter of different types of fibers. In this study, the OFDA was evaluated for measuring the ribbon width of cotton fiber snippets. One hundred and four carded cotton samples covering a wide range of fiber properties were selected and tested with nine replications per sample. The repeatability of the OFDA diameter measurement was confirmed. Comparisons with the data (cross-sections, AFIS, and Cottonscope) obtained from previous studies on the 104 samples were used to assess the effectiveness of the proposed method. The relationship between ribbon widths measured by Cottonscope and OFDA was found to be weak. Yet, there is a strong correlation between OFDA ribbon width and AFIS standard fineness. We hypothesize that the poor relationship between the Cottonscope and the OFDA is related to the medium in which the measurements are performed. The results are stable and provide measurements closely related to AFIS standard fineness, a trusted measurement within the Textile Industry.

摘要

现有的测量细度的方法耗时、繁琐，并且无法评估许多样品. 因此，纺织工业一直在寻求一种快速、准确、可靠的仪器来测定纤维细度. 最近，光纤直径分析仪（OFDA）被引入，用于快速准确地评估不同类型光纤的直径. 在这项研究中，对OFDA进行了评估，以测量棉纤维片段的带状宽度. 选择了104个涵盖广泛纤维特性的梳棉样品，并对每个样品进行了9次重复测试. 确认了OFDA直径测量的可重复性。与之前对104个样本的研究中获得的数据（横截面、AFIS和Cottonscope）进行比较，以评估所提出方法的有效性. Cottonscope和OFDA测量的色带宽度之间的关系很弱。然而，OFDA色带宽度与AFIS标准细度之间存在很强的相关性. 我们假设Cottonscope和OFDA之间关系不佳与进行测量的介质有关. 结果稳定，提供了与AFIS标准细度密切相关的测量值，这是纺织行业内值得信赖的测量值.

KEYWORDS

Cotton fiber fineness; ribbon width; fiber diameter; fiber cross-sections; optical fiber diameter analyzer; fiber quality

关键词

棉纤维细度; 色带宽度; 纤维直径; 纤维横截面; 光纤直径分析仪; 纤维质量

Introduction

Cotton fibers are single elongated cells initiated on the cotton seed's outer epidermis where the perimeter of fiber is fixed, and the cell wall fills in through cellulose deposition (maturity). When the ball opens, the fibers collapse, and the smooth fibers turn into various shapes of flat to kidney cross

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sections with convolution and variation along their length (Berkley 1948; Dochia et al. 2012). This attribute will contribute to a variation in cotton fiber diameter or ribbon width.

The cotton fiber "fineness" is often defined in terms of biological or gravimetric fineness, as an essential characteristic of cotton to process fibers into yarns. Biological fineness describes the diameter or outer perimeter of the cross-section of cotton fiber and gravimetric fineness refers to the mass of cotton per unit length of fiber known as the linear density of the fiber (Delhom, Kelly, and Martin 2018; Hequet et al. 2006; Ramey 1982; Rodgers et al. 2013). Fibers that are long, strong, and fine allow the production of higher yarn counts, i.e., finer and stronger yarns, due to the higher number of fibers in each cross-section of yarn (Delhom, Kelly, and Martin 2018; Dochia et al. 2012; Kelly et al. 2015; Tesema et al. 2023).

Several direct and indirect methods are available to evaluate cotton fiber fineness (Bragg and Shofner 1993; Brown and Graham 1950; Delhom, Kelly, and Martin 2018; Hequet et al. 2006; Kim et al. 2014; Liu et al. 2019; Paudel, Hequet, and Abidi 2013; Ramey 1982; Rodgers et al. 2015; Xu and Huang 2004). The Cottonscope and the Advanced Fiber Information System (AFIS) are currently the most common tools for determining the fineness of cotton fibers. In addition, the cross-section method, while tedious and time-consuming, is the current reference method for measuring maturity, from which we can obtain the fiber circumference directly from the image of the cross-sections. In fact, maturity is calculated from the cell wall area and perimeter (Hequet et al. 2006).

In the cross-section method, a bundle of cotton fibers is embedded in a polymer to hold the fibers together and cut into 1.0 μm-thick cross sections, which are then mounted onto a glass slide. The fiber cross-sections are then imaged by a digital microscope and analyzed using a software to measure the circumference and the area of the cell wall. This is a direct measurement method that is used for calibrating and validating indirect methods (Hequet et al. 2006; Matic-Leigh and Cauthen 1994; Rajasekaran and Thibodeaux 2002; Xu and Huang 2004). As previously mentioned, the sample preparation is tedious and requires a skilled operator.

The Cottonscope was designed to measure the fineness and maturity of cotton fibers in a relatively short period of time. It can measure 20,000 fiber snippets in about 30–40 s. It uses polarized light microscopy and image analysis software to measure the ribbon width of 0.7 mm-long cotton fiber snippets that are dispersed in water (Delhom, Kelly, and Martin 2018; Kim et al. 2020; Rodgers et al. 2013, 2015). Previous studies have shown that in water, cotton fibers swell and change their cross-sectional area by about 30% compared to their dry state (Moore et al. 1950; Ottesen and Syverud 2021).

In the case of AFIS, a 30 cm hand-crafted sliver prepared from a 0.5 g of raw cotton is used for testing. Fibers are individualized through the inbuilt opening system and then presented to electro-optical sensors to provide the length, maturity, and fineness of the cotton sample. It can measure approximately 10,000 individual fibers in about 5.0 min (Bragg and Shofner 1993; Kelly and Hequet 2018; Kelly et al. 2015) and provides distributions of the measured fiber qualities along with several statistics calculated from those distributions. The ratio of fineness to maturity is termed as standard fineness, which gives an estimate of cotton's diameter (Hequet et al. 2006; Lord 1956). The relationship between the outer perimeter and the standard fineness of cotton fibers, which could be used to estimate cotton fiber diameter, was established on a set of 104 samples by Hequet and his coauthors (Hequet et al. 2006). Nevertheless, the sample preparation is laborious and requires skilled operators.

In general, the existing methods are time-consuming, tedious, and impractical for measuring large number of commercial and research samples. For this reason, the textile industry has sought a rapid, accurate, and reliable instrument to determine fiber fineness/diameter and other fiber quality parameters.

Recently, an Optical Fiber Diameter Analyzer (OFDA) was introduced as a rapid, accurate, and reliable tool to evaluate the diameter of different types of fibers, i.e., wool, cotton, and synthetic fibers. It was developed in the 1990s by BSC Electronics Pty Ltd. to measure the diameter of wool fibers. Since then, several improvements have allowed it to measure both diameter and length (Baxter, Brims, and Taylor 1992; Brims 2002; van der Sluijs, Delhom, and Martin 2021). The latest version capable of measuring both diameter and length is called the OFDA 4000.

The OFDA 4000 has two main operating modes (i.e., slide mode and top or tuft mode) to evaluate the diameter and length of different types of fibers. The slide testing mode is used to measure only fiber diameter, while the top or tuft testing mode is capable of measuring both the diameter and length of the fiber. In this instrument, the measuring system incorporates a digital microscope with a light-emitting diode (LED) backlight source and a CCD video camera. The CCD video camera captures video images of individual fibers that are processed and analyzed by a personal computer attached to the instrument. As the AFIS, the OFDA 4000 produces a distribution of both length and diameter of a sample of fibers from which other statistical parameters are calculated.

The utility of the OFDA for determining wool fiber diameter has been demonstrated in prior studies (Baxter, Brims, and Taylor 1992; Brims 2002; Rafat et al. 2007; van der Sluijs, Delhom, and Martin 2021). However, information on using the OFDA 4000 for other natural fibers is limited. OFDA 4000 is hypothesized to be a valuable tool for evaluating the diameter of various textile fibers, including cotton, hemp, flax, jute, and synthetic fibers. This study was designed to assess the accuracy and reliability of the OFDA 4000 to measure cotton fiber ribbon width. The term ribbon width is used to describe the 2D projected image of cotton fiber, which is expected to be a good estimate of diameter. In general, the width of the ribbon shall be as wide as the diameter if the object has a cylindrical tube structure. Previous studies revealed that cotton fibers develop as cylindrical tubes and have a circular cross-section in the boll just before they lose their moisture (Peeters et al. 1986). However, when cotton is dried naturally, the tube structure collapses and forms a flat ribbon-like structure of uneven diameter or irregular cross-sections with recurring convolutions or twists along the length of the fiber (Duckett and Cheng 1972; Hequet et al. 2006; Hsieh 2006; Wang and Wang 2009). Mature and immature cotton fibers will behave differently. Cotton fiber generally shows a bean-shaped cross-section after drying (Hequet et al. 2006; Hsieh 2006; Kelly et al. 2015). As a result, the two measurements are not exactly equivalent, but it is possible to estimate the diameter of cotton fiber from its ribbon width (Ramey 1982).

Materials and methods

Two independent sample sets were used in a two-phase experiment. The first set aimed to examine the stability of fiber diameter measurements over several days using the OFDA 4000. The second set focused on measuring the width of cotton fiber ribbons and assessing the agreement of the OFDA 4000 with other instruments.

For the stability study, two premade fiber snippet slides representing fine and coarse polyester fibers were used. These premade slides were provided by the OFDA 4000 manufacturer to adjust the camera focus when an "Out of focus" message appeared on the computer screen and to evaluate the performance of the instrument. These slides were measured with the OFDA 4000 snippet testing mode using a research protocol of 3 readings per sample per day for 18 days. Regression analysis was performed to assess the presence of trends (i.e., machine drift), and the CV% among daily measurements was computed and used as a standardized method for measuring the variation of the diameter measurements. Another set of 104 samples covering a wide range of cotton fiber properties was selected for the second experiment (Tables 1 and 2).

Table 1. High volume instrument fiber properties of 104 samples.

Parameters	Mic	UHML [in.]	UI [%]	Str [g/tex]	Elo [%]	Rd	+b
Average	4.22	1.08	81.87	29.40	5.38	77.02	10.81
Min	2.60	0.96	78.59	21.58	3.59	64.13	8.36
Max	5.65	1.31	84.53	41.30	8.28	81.07	15.80
Range	3.05	0.35	5.94	19.72	4.69	16.94	7.44
Std. Dev.	0.56	0.07	1.19	3.21	1.03	3.02	1.46
CV%	13.17	6.54	1.45	10.93	19.18	3.93	13.50

Table 2. Advanced fiber information system fiber properties of 104 samples.

Parameters	UQL(w) [in]	L(n) [in]	SFC(n) [%]	Fine [mTex]	Mat Ratio	Hs
Average	1.12	0.74	28.29	166.23	0.87	191.74
Min	0.97	0.59	19.08	143.70	0.74	164.04
Max	1.38	0.93	43.04	194.50	0.95	208.31
Range	0.42	0.34	23.96	50.80	0.21	44.28
Std. Dev.	0.09	0.06	4.66	9.35	0.04	9.50
CV%	7.60	8.38	16.47	5.62	4.27	4.95

The 104 samples are known cotton samples and were collected to establish a reference material for cotton maturity. These samples contain two major cultivated cotton species viz. *Gossypium hirsutum* and *Gossypium barbadense*. Most of the samples were collected from cotton grown in the USA, and the rest were from six foreign countries, namely Egypt, Uzbekistan, Pakistan, Cameron, Benin, and Australia (Hequet et al. 2006). These samples were conditioned for 2 days before testing, at a standard temperature of 21 ± 1°C and a relative humidity of 65 ± 2%.

Sample preparation and testing procedures

From each sample, about 15 mg of cleaned cotton was randomly selected and placed onto a guillotine to obtain 2.0 mm fiber snippets. The fiber snippets, nearly 10 mg by weight, were placed in a spreader to evenly distribute them onto a 70 mm glass slide (Figure 1). After the necessary operational setting was performed, the glass slide with fiber snippets was attached. The fiber snippets were then scanned and photographed using a digital microscope with an LED light source and a CCD video camera (Figure 2).

The images were processed and analyzed using OFDA software provided by the manufacturer. Each sample was tested on a research protocol of nine replications, and more than 10,000 fiber snippets per slide, or replication, were measured within 40 s. The average of nine replications for each sample was compared to other instruments or methods, such as the cross-section method, AFIS, and Cottonscope. The coefficient of determination (R^2) was obtained for each relationship to determine the quantity of variance. In general, a higher R^2 indicates that the regression model explains more variability.

Results and discussion

Diameter stability study

The stability of the diameter measurements of the OFDA 4000 was evaluated with the two premade slide samples. The three measurements for each premade slide with the polyester fiber snippets were averaged daily for 18 days to detect a possible measurement trend or machine drift. Figure 3 shows the average diameter for each day in a scatter plot, while Figure 4 illustrates the complete distributions of diameter measurements over the study period to identify possible trends.

The diameter measurements remain consistent, with no noticeable upward or downward trend (Figure 3). In addition, as Figure 4 illustrates, similar shapes with slight deviations were observed in the distributions of diameter measurements over time. To validate this, we conducted a simple linear regression analysis on the complete set of diameter measurements for each sample. The results revealed that the slopes are not significantly different from zero at a 95% confidence level (i.e., Slope 1 = −0.0021 ± 0.0023 and Slope 2 = −0.0025 ± 0.0035 for Samples 1 and 2, respectively), indicating no significant upward or downward trend on the diameter measurement over time. In addition, the coefficients of variation among the 18 diameter measurements were computed for each sample to measure the variation of the measurements. The CV% of the average diameter across days was

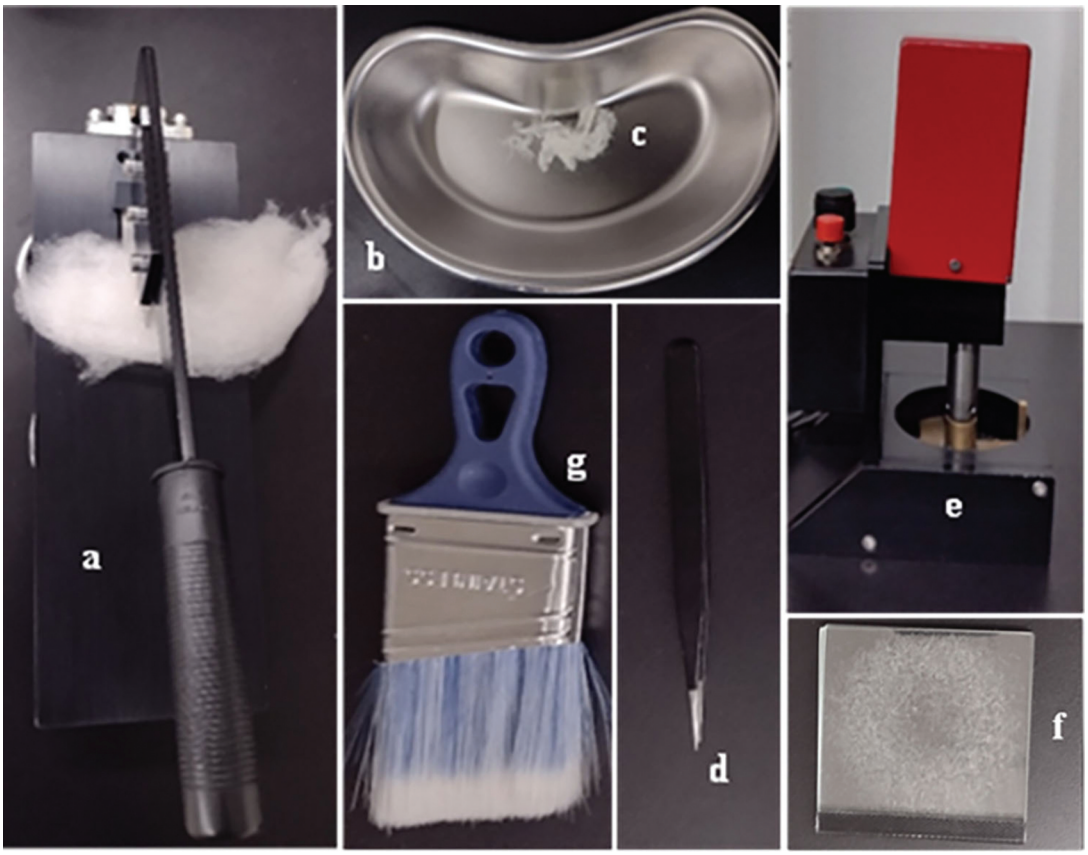


Figure 1. Equipment used in sample preparation to measure cotton fiber ribbon width: (a) guillotine, (b) snippets dish, (c) fiber snippets, (d) blunt nose tweezers, (e) spreader, (f) fiber snippets slide, and (g) brush.

extremely low in both sets of data (i.e., 0.13% for Sample 1 and 0.14% for Sample 2). Consequently, the diameter measurements remain consistent and reliable throughout the duration of the study.

OFDA 4000 ribbon width measurement of cotton fiber

The study on measuring the width of cotton fiber ribbons was conducted after the reliability of OFDA diameter measurements was confirmed. Each sample was tested following a research protocol of nine replications. The average ribbon width of each replication was acquired during the standard measurement process. A complete ribbon width distribution of each replication per sample was obtained from the system with a few additional steps. This distribution provides a more detailed understanding of the variations in ribbon width within each sample. For each sample, the average ribbon width measurement of nine replications was computed to obtain a representative ribbon width for that sample.

Figure 5 shows a complete distribution of ribbon widths for the set of 104 samples. Collectively, these distributions demonstrate a broad range of ribbon widths, as is the case for other fiber properties with this set of samples.

The average ribbon width was between 11.14 µm and 13.87 µm. It was found that the coefficient of variation was extremely low across the nine replications of each sample and varied from 0.35% to 1.53%, as expected in well-blended and carded sample. In addition, it was noticed that the quantity of measurements in each replication ranged from 15,323 to 22,206. Using these data, we established relationships between the OFDA 4000 ribbon width

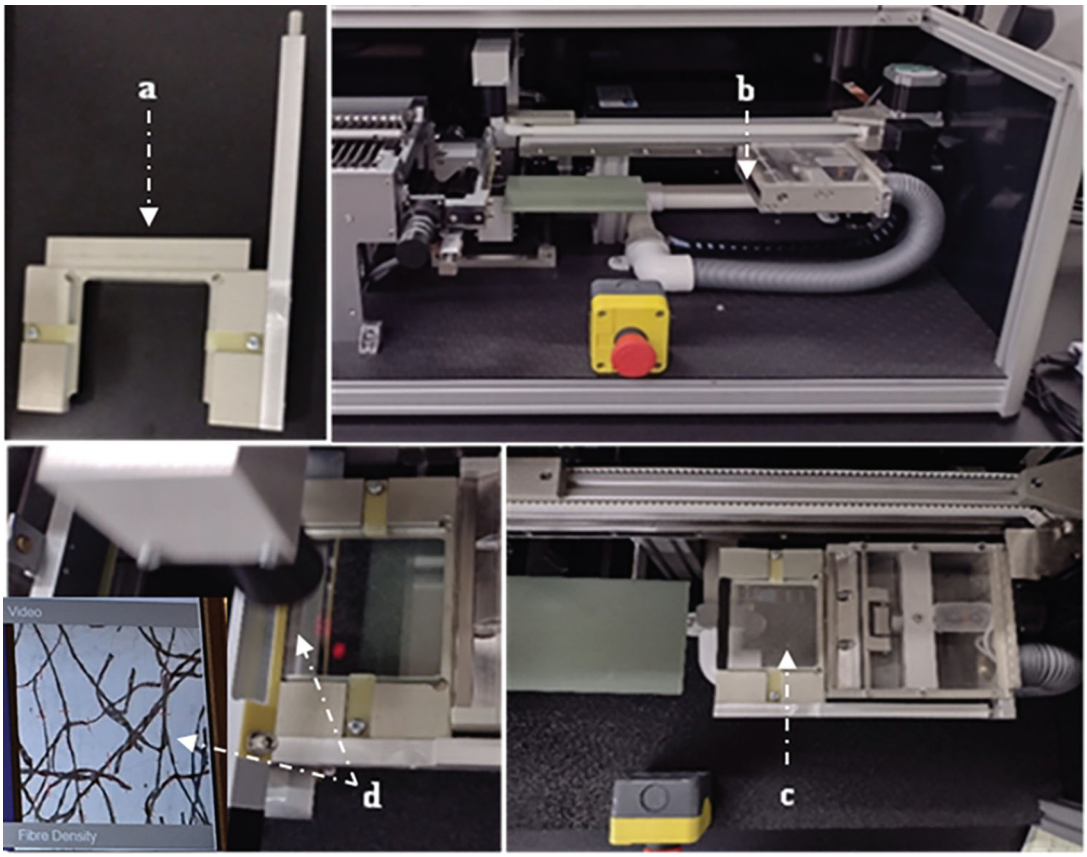


Figure 2. The essential components of the OFDA 4000 for measuring cotton fiber ribbon width: (a) 70 mm slide holder, (b) gripper, (c) 70 mm slide holder and fiber snippets slide, and (d) measuring system (digital microscope and CCD camera).

and results from existing methods for the same or similar fiber characteristics. For comparison, the data (cross-sections, AFIS, and Cottonscope) obtained from past studies on the 104 samples were used to assess the effectiveness of the OFDA 4000 ribbon width measurement on cotton.

Relationship between cottonscope ribbon width and OFDA 4000 ribbon width

Both the Cottonscope and the OFDA 4000 measure cotton ribbon width from several fiber snippets. Yet, the measurements are performed in two different mediums, i.e., aqueous medium for the Cottonscope and air for the OFDA 4000. Figure 6 shows the relationship between ribbon widths from the Cottonscope and the OFDA 4000.

There is a weak positive correlation between Cottonscope ribbon widths and OFDA 4000 ribbon widths for the set of 104 samples. Only 36% of the variation can be explained with this relationship ($R^2 = 0.36$). We also noticed that the Cottonscope measurements were higher than those of the OFDA 4000. It is known from earlier studies that cotton swells in water and changes its cross-sections substantially (Moore et al. 1950; Ottesen and Syverud 2021). As such, we hypothesize that the poor relationship between the Cottonscope and the OFDA 4000 is related to the difference in the medium in which the measurements are performed. It is important to note that these measurements are carried out in two different states viz. wet and dry.

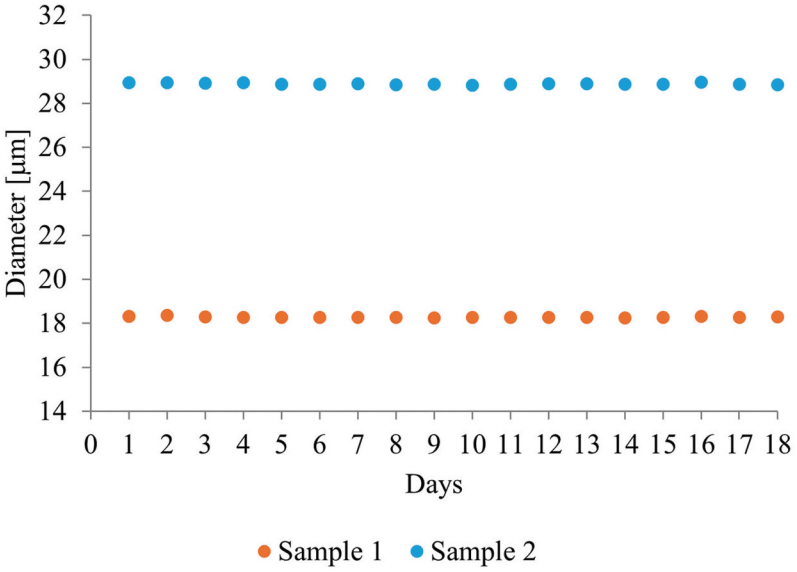


Figure 3. Repeatability of the OFDA 4000 diameter measurement.

Relationship between cross-section diameter and OFDA 4000 ribbon width

As with the Cottonscope, the correlation between cross-sectional diameter and the OFDA 4000 ribbon width was weak (Figure 7) likely for similar reasons.

Each method employs a distinct procedure for preparing samples. In the sample preparation process for the cross-section method, fibers are embedded in a polymer and undergo a treatment prior to measurement. As such, this process likely causes fiber swelling to some degree.

Relationship between AFIS standard fineness and OFDA 4000 ribbon width

Both the AFIS fineness and OFDA ribbon width measurements are performed on similar medium and raw cotton samples. However, they use different sample preparation methods and samples, i.e., 30 mm-long hand-crafted sliver for the AFIS and 2.0 mm-long fiber snippets for the OFDA. In addition, the AFIS data is obtained on complete fibers (i.e., 3000 fibers per replication). The relationship between AFIS standard fineness (fineness/maturity ratio) and OFDA ribbon widths is shown in Figure 8.

The AFIS standard fineness and the OFDA 4000 ribbon width are strongly correlated ($R^2 = 0.81$). This means that the AFIS standard fineness can explain more than 80% of the variation in OFDA ribbon width. As these methods use relatively different sample preparation and measurement procedures, we do not expect a perfect correlation.

Conclusions

The Optical Fiber Diameter Analyzer (OFDA) is an instrument that is marketed as a multifiber testing instrument. Until now, there has been no data published on the efficacy of this instrument for measuring the fineness or ribbon width of cotton fibers.

This study was designed to investigate the repeatability of the OFDA 4000 diameter measurements, assess the ribbon width of cotton fibers, and explore the relationship between OFDA ribbon width and comparable outputs from existing methods. Over a four-week period, 18 tests were conducted to assess the stability of OFDA diameter measurements. For each sample, simple linear regression was applied

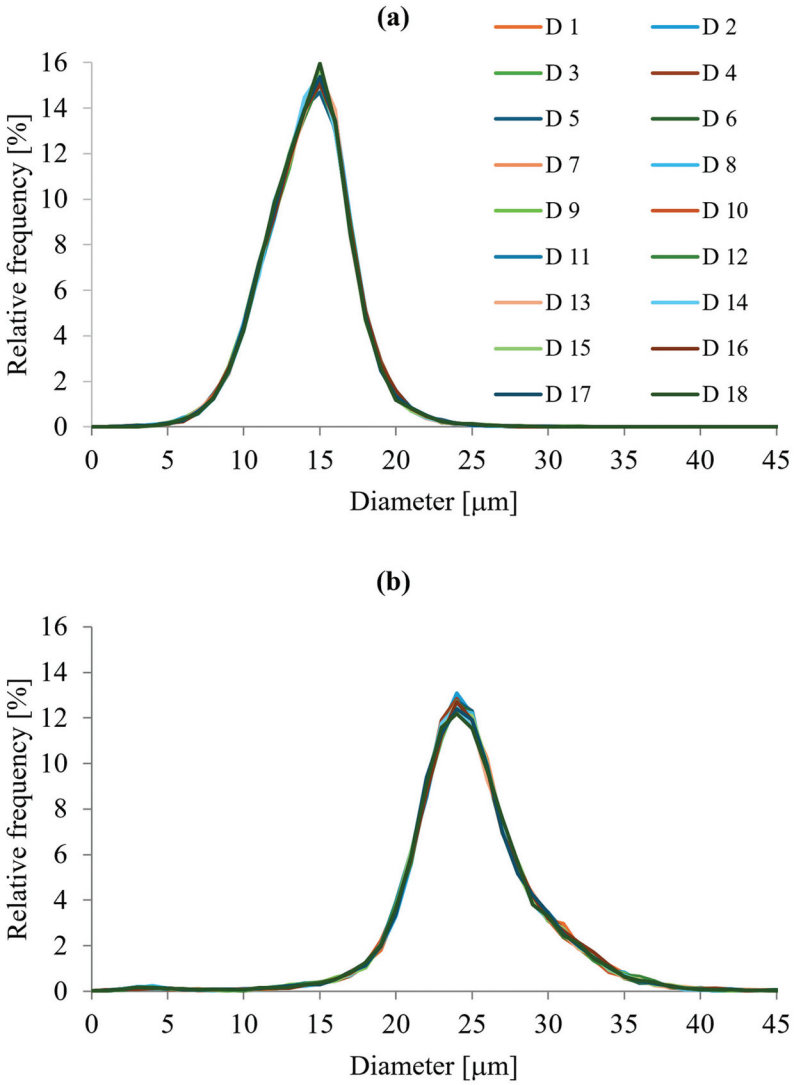


Figure 4. The OFDA 4000 diameter measurement distributions for (a) sample 1 and (b) sample 2.

to the collected data. The slope and offset values were analyzed to identify any trends that might indicate machine drift. Ideally, a slope of zero would suggest no drift, with the best-fit line being horizontal and the measured values randomly distributed around it. The linear regression analysis indicated that the slopes were not significantly different from zero at a 95% confidence interval, with values of −0.0021 ± 0.0023 for Sample 1 and −0.0025 ± 0.0035 for Sample 2. This implies no significant upward or downward trend in the measurements over time. In addition, the coefficients of variation for the 18 diameter measurements in each sample were calculated to assess measurement variability. Both samples showed very low variation, with 0.13% for Sample 1 and 0.14% for Sample 2. These findings suggest that the diameter measurements remained consistent and reliable throughout the study period.

The cotton fiber ribbon width was measured on a set of 104 samples, which encompassed a broad range of fiber properties. This measurement was conducted after confirming the repeatability of OFDA diameter measurements. The average ribbon width ranged from 11.14 μm to 13.87 μm. Moreover, the coefficient of variation across the nine replications per sample was relatively low,

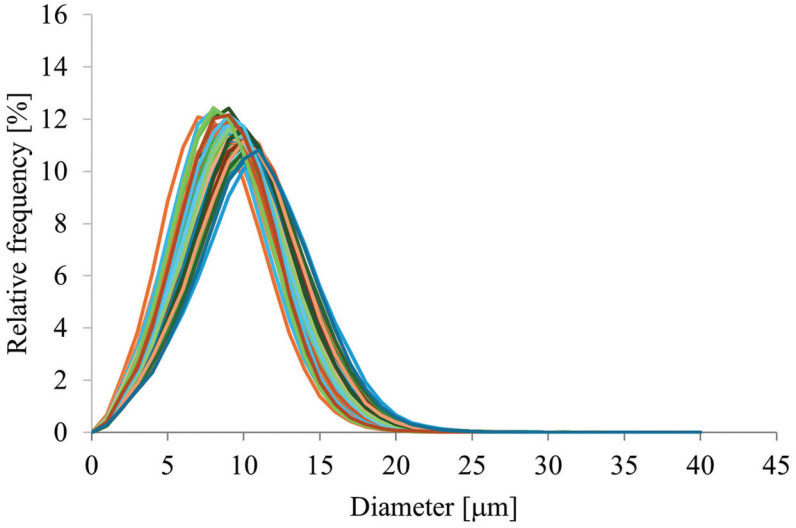


Figure 5. Cotton fiber ribbon width distributions for the set of 104 samples.

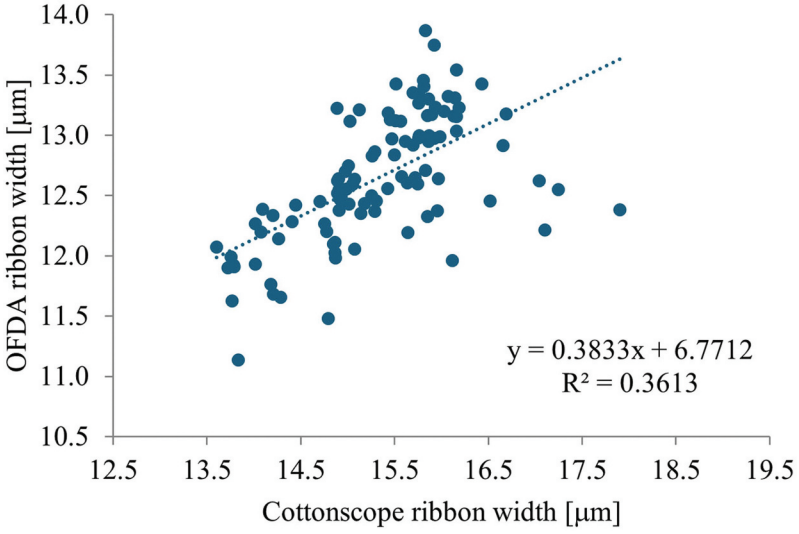


Figure 6. The relationship between Cottonscope ribbon width (CS_RW) and OFDA ribbon width (OFDA_RW).

between 0.35% and 1.53%, which is typical of a well-blended and carded sample. The reliability of the OFDA was assessed by establishing relationships between the ribbon width of OFDA 4000 and the outcomes of existing methods for similar fiber characteristics. Nevertheless, when compared with other instruments that either measure ribbon width directly or measure other fiber properties closely related to ribbon width, we observed mixed results. For example, the OFDA ribbon width does not correlate well with the Cottonscope ribbon width nor the cross-section diameter. We hypothesize that these poor relationships (i.e., OFDA 4000 versus Cottonscope and OFDA 4000 versus cross-section) have something to do with the medium used for measuring. With the Cottonscope and the cross-section method, fibers are immersed in an aqueous medium. On the other hand, a strong positive correlation was observed between the ribbon width measured by the OFDA 4000 instrument and the standard fineness determined by the AFIS, which makes sense because both measurements are

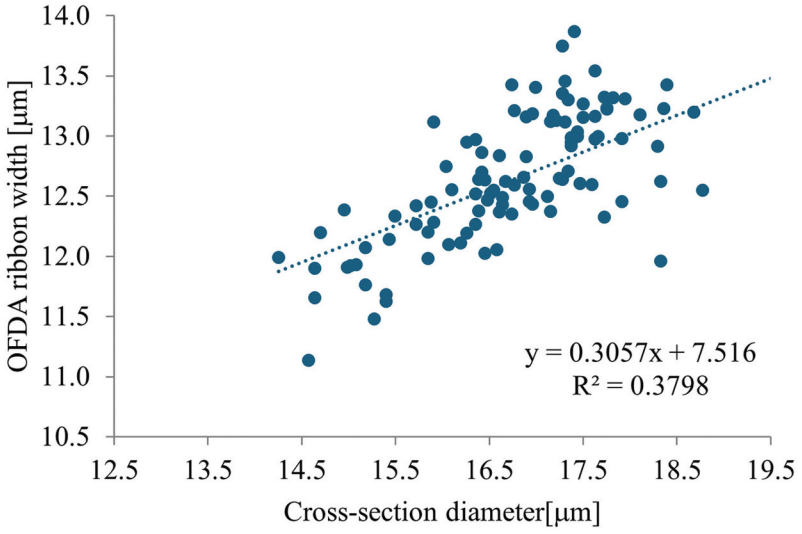


Figure 7. The relationship between cross-sectional diameter (Cs_dia) and OFDA 4000 ribbon width (OFDA_RW).

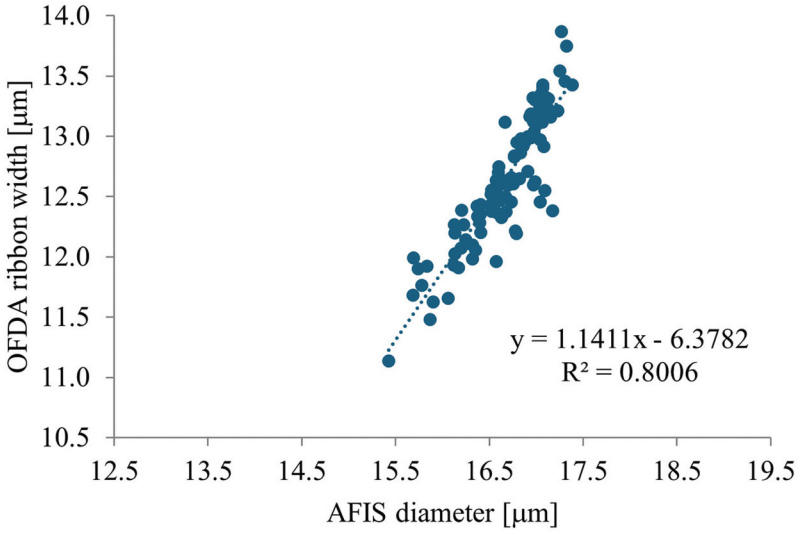


Figure 8. The relationship between AFIS standard fineness (AFIS Hs) and OFDA ribbon width (OFDA_RW).

conducted on fibers in a normal, dry state. The AFIS standard fineness accounts for over 80% of the variation in OFDA ribbon width.

In general, the OFDA 4000 has proven to be a reliable instrument for measuring the ribbon width of cotton fibers. It is particularly useful for studying cotton fiber development due to its minimal sample size requirement and nondestructive testing method, which prevents fiber breakage. The results are stable and provide measurements that are closely related to AFIS standard fineness, which is a trusted measurement within the Textile Industry. Even though the results obtained are promising, they need to be validated with a large set of commercial bales.

Highlights

- An Optical Fiber Diameter Analyzer (OFDA) was introduced as a reliable and potential tool for measuring ribbon widths of cotton fiber.
- The repeatability of the OFDA diameter measurement was confirmed before the ribbon width measurement.
- Comparisons with the data using existing methods obtained from previous studies were used to assess the effectiveness of the proposed method.
- OFDA ribbon width measurements are closely related to AFIS standard fineness and could be useful to determine fiber quality.

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Disclosure statement

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